CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (Mechanical, CAD/CAM)

November 2018

ME751/ME701.01 Computer Aided Design

Date: 29.11.2018, Thursday **Time: 1:30 p.m. to 4:30 p.m.** **Maximum Marks: 70**

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of scientific calculator & PSG data book is allowed.

SECTION- I

Q1 (a) Using a simplified description of process of design explain where computational aids might be assistance to the designer. **5**

(b) Answer following with correct option. **6**

1. C^1 continuity refers to:
   (A) Common tangent (C) Common curvature
   (B) Common point (D) Common normal
2. The degree of the Bezier curve with n control points is:
   (A) n + 1 (C) n - 1
   (B) n (D) 2n
3. Which attributes of image transformation rotate the image by a given angle?
   (A) Rotate - X (C) Both (A) & (B)
   (B) Rotate - Y (D) None of these
4. Bresanham circle algorithm uses the approach:
   (A) Midpoint (C) Point
   (B) Line (D) None of these
5. Types of computer graphics are:
   (A) Vector and raster (C) Vector and scalar
   (B) Scalar and raster (D) None of these
6. In a display specified as 600 x 400 the number of pixels across the display screen is:
   (A) 600 (C) 400
   (B) 240000 (D) Both (A) & (B)

Q2 (a) Differentiate between World Coordinate System (WCS) and User Coordinate System (WCS). **3**

(b) Explain the working principle of plotter. **3**

(c) Using Bresenham's line algorithm, find the pixel positions between end points (2, 7) and (15, 10). **6**

OR

Q2 (a) Explain the types of coordinate systems used in geometric modeling. **3**

(b) State difference between raster scan and vector scan display. **3**

(c) Explain generation of circle using Bresenham's Midpoint algorithm. **6**

Q3 (a) Compare geometric transformation with coordinate transformation. Also explain composite transformation. **3**

(b) What are the advantages of homogenous coordinate system? **3**

(c) Perform 45 degree CW rotation of a triangle having coordinates A (0, 0), B (3, 2) and C (2, 3) about origin. Find the new vertex positions of triangle. **6**

OR

Q3 (a) What is concatenated transformation? Explain its benefits. **3**

(b) Explain 3D rotation transformations. Also give transformation matrix for each. **3**

(c) A triangle with vertices A (5, 5), B (8, 5) and C (5, 10) is to be reflected about the line y = 3x + 4. Determine the composite transformation matrix. **6**

SECTION- II

Q4 (a) Write down the steps for generation of solid model of a square bolt using any solid modeling software. **4**

(b) What is parametric form of an equation? Develop the parametric equation for circle. **4**

(c) Explain the terms "geometry" and "topology" in solid modeling? **3**

Q5 (a) What is reverse engineering? Explain in brief different type of scanners? **6**

(b) A Cubic Bezier Curve is defined by points (1,1) (2,3) (4,4) & (6,1). Calculate the coordinate of parametric midpoint of the curve & verify that its gradient (dy/dx) is $1/7^{th}$ at this point. **6**

OR

Q5 (a) Describe the concept of feature based modeling. **4**

(b) Explain the various techniques used for surface modeling. **4**

(c) Calculate the parametric points of Hermite cubic curve that fits the points P_0 = (0, 3), P_1 = (4, 2) and the tangent vectors P'_0 = (0.707, 0.707), P'_1 = (0, 1) at u = 0.25. **4**

Q6 (a) Write a short notes on the following: **6**

1) IGES
2) GKS

(b) Write an interactive algorithm for design of spring. **6**

OR

Q6 (a) Discuss the need for standardization in computer graphics & list the various CAD standards. **6**

(b) Prepare an exhaustive algorithm for design of connecting rod. **6**

*****